

# Probability for NLP

CSCI 3495: Natural Language Processing & Large Language Models • W4C1

## 1. Probability

Probability measures how likely a given event is, expressed as a number from 0 (impossible) to 1 (certain). An *event* is a thing that happens; an *outcome* is what it produces. Write  $C(A)$  for the number of times  $A$  happened and  $N$  for the number of trials.

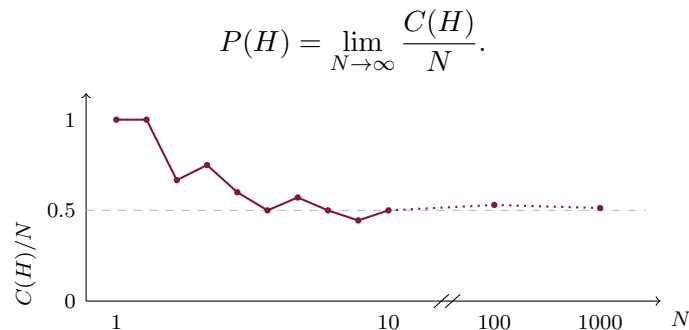
A tossed coin has two outcomes, heads  $H$  and tails  $T$ . They are complements, so one number fixes the other:

$$P(H) = 1 - P(T), \quad P(H) \approx P(T) \approx 0.5.$$

**How do we know this?** Empirical evidence. Toss the coin  $N$  times and count. Ten tosses came out  $HHTHTTHTTH$ , which accumulates to the counts below; the last two columns continue the same run.

| $N$      | 1    | 2    | 3    | 4    | 5    | 6    | 7    | 8    | 9    | 10   | 100  | 1000 |
|----------|------|------|------|------|------|------|------|------|------|------|------|------|
| $C(H)$   | 1    | 2    | 2    | 3    | 3    | 3    | 4    | 4    | 4    | 5    | 53   | 513  |
| $C(T)$   | 0    | 0    | 1    | 1    | 2    | 3    | 3    | 4    | 5    | 5    | 47   | 487  |
| $C(H)/N$ | 1.00 | 1.00 | 0.67 | 0.75 | 0.60 | 0.50 | 0.57 | 0.50 | 0.44 | 0.50 | 0.53 | 0.51 |

Every row is a count, and the estimate in the last one is a count divided by a count. It is noisy while  $N$  is small and settles as  $N$  grows:



As the number of trials increases, the true probabilities of the events emerge, assuming no time dependence.

## 2. Sequential vs. independent events

Two coin tosses,  $t_1$  and  $t_2$ . What is the probability that  $t_1 \rightarrow H$  and  $t_2 \rightarrow H$ ? The chain rule splits the joint into one toss and then the other:

$$P(H_2 \cap H_1) = P(H_2 | H_1) \cdot P(H_1).$$

So the joint needs the conditional. What is the probability of  $t_2 \rightarrow H$  if  $t_1$  was  $H$ ? The coin has no memory, so knowing  $H_1$  says nothing about  $H_2$ :

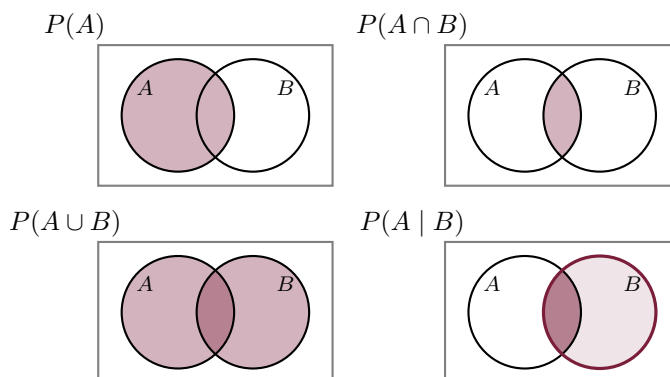
$$P(H_2 | H_1) = P(H_2) = 50\%.$$

That is what independence means, and substituting it back finishes the joint:

$$P(H_2 \cap H_1) = P(H_2) \cdot P(H_1) = 0.5 \cdot 0.5 = 25\%.$$

### 3. Venn diagram notation

Let  $A$  and  $B$  be events inside a sample space.



$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \quad (1)$$

$$P(A | B) = \frac{P(A \cap B)}{P(B)} \quad (2)$$

$$P(A \cap B) = P(A | B) \cdot P(B) \stackrel{\text{independent}}{=} P(A) \cdot P(B) \quad (3)$$

Line (1) subtracts the overlap because adding the two circles counts it twice. Line (2) shrinks the sample space to  $B$ : the denominator is no longer the whole box. Line (3) is (2) rearranged, and only collapses to a product when conditioning on  $B$  tells you nothing about  $A$ .

Splitting  $B$  over  $A$  and its complement gives the law of total probability:

$$P(B) = P(B | A) \cdot P(A) + P(B | \neg A) \cdot P(\neg A).$$

### 4. Bayes' rule

$$P(A | B) = \frac{P(B | A) \cdot P(A)}{P(B)}$$

**How did we get this?** Start from the conditional and substitute the chain rule into its numerator:

$$P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B | A) \cdot P(A)}{P(B)}.$$

Bayes flips a conditional: it gets you from the direction you can count to the direction you want.

**Naive Bayes.** Classify by picking the class  $C_k$  that maximises

$$P(C_k | x_1, \dots, x_n) \propto P(C_k) \prod_{i=1}^n P(x_i | C_k),$$

which is Bayes' rule plus one assumption: the features are independent *given the class*,  $P(x_i | C_k, x_j) = P(x_i | C_k)$ , so line (3) turns the joint  $P(x_1, \dots, x_n | C_k)$  into a product. The denominator  $P(x_1, \dots, x_n)$  is the same for every class, so it drops out of the comparison. That is the “naive” part, and it is usually false; it survives because picking the largest score only needs the right ordering.

## 5. Exercise

Twenty app reviews, counted. Every answer below comes out of the four inner cells.

|          | has “crashes” | no “crashes” | total |
|----------|---------------|--------------|-------|
| 1 star   | 7             | 3            | 10    |
| 4–5 star | 1             | 9            | 10    |
| total    | 8             | 12           | 20    |

1.  $P(1 \text{ star})$ ,  $P(\text{crashes})$ ,  $P(\text{crashes} \cap 1 \text{ star})$
2.  $P(\text{crashes} \cup 1 \text{ star})$
3.  $P(1 \text{ star} \mid \text{crashes})$  and  $P(\text{crashes} \mid 1 \text{ star})$
4. Is  $P(\text{crashes} \cap 1 \text{ star}) = P(\text{crashes}) \cdot P(1 \text{ star})$ ?
5. “update” is in 10 reviews, five of them 1 star. Same question.
6. A flagger catches 80% of 1 star reviews and 20% of the rest. Find  $P(1 \text{ star} \mid \text{flagged})$ .